Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

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***Abstract*—Location-based social networks (LBSNs) record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can act ahead of demand. Two coarse questions about the next visit are usually enough: its category (the kind of place) and its region (which part of the city). These are normally handled by separate models. We therefore ask whether one model can learn both, and what sharing a single representation costs. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across six datasets, five U.S. states and one non-U.S. city (Istanbul), this lifts next-category prediction by a wide margin over a standard place embedding [1] (about $+28$ to $+40$ macro-averaged F1), and most of the gain comes from the per-visit context. We then train one multi-task model for both tasks. At every dataset it outperforms a dedicated category model (about $+5$ to $+9$ macro-F1), and on region it outperforms the dedicated region model at four of the six datasets, while matching it (statistically, within two points) at the other two. Across the five U.S. states the gain on region grows with the number of regions, with the two largest states still single-seed. On the non-U.S. city the result holds: the joint model outperforms on category and comes out slightly ahead on region.**

***Index Terms*—next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks, mobility data**

I. Introduction

Location-based social networks, such as Gowalla [2], let people share the places they visit. Each check-in is a small record of human movement, and together these records show how people move through a city. A natural and useful question is what a person will do next. Individual mobility is highly predictable in principle [3], and a service that can anticipate the next move can prepare ahead of time instead of reacting after the fact. For a mobility-aware service this can mean staging the right content in the area a user is heading to [4], or planning capacity there before demand arrives. The same logic is already established at the network level, where predicted handovers let cellular services adapt in advance [5]; at the city scale, check-in mobility analysis is likewise read from past traces and treated as a design input for mobility-aware services [6]. We ask the predictive, city-scale version of the question.

Predicting the next place exactly, a single point of interest (POI), is hard, and it is often more than a service needs. Two coarser (less detailed) questions are easier to answer and usually enough: the next category, the kind of place such as food or shopping, and the next region, the part of the city a user is heading to (a neighborhood-scale unit, the census tract in the U.S. and the *mahalle* in Istanbul). These two questions pair naturally, one for intent and one for geography, and they are easier to predict accurately than a single POI while still being useful. The question we study is whether one model should learn both at once.

Sharing one representation across tasks is not free. Joint training can help one task while hurting the other, a well-known trade-off in multi-task learning (MTL): one model does several jobs at once by sharing most of its parts, and as a result the shared parameters settle on a compromise that suits neither task perfectly [7]. Our earlier work [8] observed exactly this for next-category and next-region. So the useful question is not whether MTL is good in general, but where sharing helps, where it costs, and how to share so the gains hold and the cost stays small.

We introduce two changes and measure each one carefully. First, we build a check-in-level representation: instead of one fixed vector per place, where two visits to the same coffee shop look identical, each check-in gets its own vector that carries its context (the time, the nearby places, and the recent trail). This extends hierarchical graph representations of places [1], and the infomax idea behind them [9], by adding a fourth, check-in level beneath the place, region, and city levels (Fig. 1). Per-visit context is not new on its own; what is new is obtaining it from inside this hierarchical graph representation, a choice we test directly (Section II). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk that carries the semantic context and a private spatial path for the region task (Fig. 2). We deliberately evaluate on two very different settings: five U.S. states of different sizes (Gowalla) and one international city (Istanbul). This lets us see whether the findings hold across small and large, U.S. and non-U.S. data.

Our contributions are the following.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit. It improves next-category prediction by a large and consistent margin over a standard place embedding, about

+28 to +40 points of macro-F1 (a balanced score that counts every category equally) across all six datasets. A controlled test shows the gain comes from the per-visit context, not from extra supervision (Table II). The margin is large because a single fixed vector per place cannot separate places that host several kinds of activity, which per-visit context recovers.
- A single model for joint next-category and next-region prediction that shares a semantic context across the two tasks while keeping a private path for the spatial one. In one forward pass it outperforms a dedicated single-task category model at every dataset (by about +5 to +9 points of macro-F1), even though each dedicated model is tuned per dataset. On region, it outperforms at four of the six datasets and stays statistically non-inferior within a two-point margin (TOST, a test of equivalence) at the remaining two (Table III).
- An empirical account, across five states of different sizes and one international city, of how the joint gain scales with the number of regions. The category gain holds everywhere, and across the U.S. states the region result moves monotonically, from a non-inferior match (TOST, ± 2 percentage points, or pp) at the small region counts to outperforming the dedicated model at the large, with the largest state (California) showing the largest region gain (Fig. 4); Istanbul, the smallest dataset, also comes out ahead on region. Four of the six datasets are measured at four seeds over five folds on both arms (the joint model and the dedicated one); California and Texas are single-seed and provisional.

The rest of the paper reviews background and related work (Section II), defines the problem and the two tasks (Section III), presents our method (Section IV) and experimental setup (Section V), reports results (Section VI), discusses findings and limitations (Section VII), and concludes (Section VIII).

II. Background and Related Work

A. From place embeddings to per-visit embeddings

A place embedding turns a POI into a vector, so that a model can reason about places by their learned geometry, where similar or nearby places sit close together. This replaces treating each place as a bare, unrelated index, an arbitrary ID number that says nothing about what the place is or where it sits. Deep Graph Infomax (DGI) [9] is a general self-supervised method for learning node representations on graphs. Applied here to a network of places, it learns such vectors by training them to tell the real place network, linked by similarity, time, and distance, from a shuffled copy. Hierarchical Graph Infomax (HGI) [1] adds geographic structure, organizing the network as place, then region, then city, so the vectors respect where places sit. Both methods produce one vector per place, so two visits to the same coffee shop, one at 8 a.m. and one at 10 p.m., look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [10] is the closest example, learning a vector per visit by hiding parts of a user's check-in sequence and learning to fill them back in. Our representation, Check2HGI, also works at the check-in level, but it gets there differently. It keeps the hierarchical place-to-region-to-city network and trains it with the same infomax objective, now extended one level deeper, from individual places down to individual check-ins, a fourth level beneath place, region, and city (Fig. 1). It does not switch to a sequence model, as CTLE does. In short, earlier place embeddings (DGI, HGI) changed *which* features a place vector encodes; CTLE and we change *how often* a vector is produced, once per place or once per visit.

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE directly to show that the combination, not per-visit context on its own, is the source of the gain.

B. Predicting the next category and the next region

A large body of work predicts the next place a user will visit, the exact POI, with recurrent and attention models such as ST-RNN [11], DeepMove [12], Flashback [13], STAN [14], and GETNext [15]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. These targets are not trivial: the region alone spans hundreds to several thousand classes. We choose them because they are what a mobility-aware service can act on, not to make the task easier. Predicting over a partition of the map is also the standard formulation in the human-mobility literature, where the target is a grid cell rather than an exact place [16]; our next-region task follows this formulation, with administrative units in place of grid cells. The category task itself follows our earlier line of work [8]. The field increasingly models several granularities at once, predicting category-level and region-level sequences alongside the place sequence; in that work, however, category and region are auxiliary signals that help a headline next-place task (SGRec [17], MCMG [18], HMT-GRN [19]). We study the pair as the object itself. To our knowledge, fine-grained region as a co-equal end target, rather than an auxiliary coarse cell, is underexplored: HMT-GRN, for instance, predicts a coarse geohash cell only as an aid to its next-place search. The nearest exceptions stop short of our pairing in one way or another: DRRGNN [20] forecasts a person's next activity region jointly with a mobility-intention label, but over regions discovered per person rather than a fixed citywide partition. A recent generative recommender [21] predicts category and region jointly, but only as auxiliary steps toward its next-place ranking. We therefore define our region unit carefully when we state the tasks.

C. Multi-task learning for mobility

Multi-task models for mobility have paired location prediction with related signals for some time. MCARNN [22] jointly predicts activity and location, iMTL [23] couples them in a shared, interactive framework built for sparse check-ins, and a recent hierarchy-aware model continues the pattern [24]; in all three the location target is the exact place. The closest alternative to our setting is the cascade, a pattern with a long lineage: predict the category first, then predict the place given the category [25], [26]. CSLSL [27] is its current form, predicting in a chain (when, then what, then where), with location as the headline output and category as an instrumental step along the way; CatDM [28] likewise uses category instrumentally, to filter candidate places. We differ in two ways. We predict category and region in parallel, as co-equal end targets in one model with a single forward pass, and we drop the next-place target entirely, so neither task is instrumental to a third. The parallel form also follows from the task pair itself: with two co-equal targets, a chain would have to promote one of them to an upstream stage whose errors the other inherits, an order nothing in the tasks dictates. Since the cascade is the pattern the field actually uses, we also test the choice directly (Section VI-B). On the optimization side we are deliberately conservative: hard parameter sharing, one shared stack of layers for both tasks, is standard practice [7], and our use of it is not itself the contribution. Rather than propose yet another gradient-balancing method (PCGrad [29], GradNorm [30], Nash-MTL [31]), we ask where sharing helps and where it costs, and we give a clear, measured account, consistent with reports that such methods rarely improve on a well-tuned fixed weighting at two tasks [32], [33]. We confirm this on our own two tasks: across the gradient-balancing methods we tried, including the three named above, none improved on a tuned fixed task weighting in our model. The reason is visible in the gradients themselves: the next-category and next-region updates barely correlate on the shared layers (a near-zero correlation, about 0.001, throughout training), so the two tasks neither pull against each other nor reinforce each other there, and a balancer has no conflict to resolve. This is consistent with the mechanism we report in Section VI-B: because the tasks do not compete on the shared layers, the category gain comes from a stronger shared trunk rather than from the region task teaching the category one. We report this as our finding for this pair of tasks, not as a general rule. The novelty here is empirical, not architectural.

III. Problem and Tasks

We work with check-in sequences. For each user we order their check-ins in time and form short windows of nine consecutive visits. From each window we predict two properties of the next visit. The first is its *category*, the kind of place, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul shares the same seven labels via the Massive-STEPS seven-root mapping. The second is its *region*, the census tract the place falls in (the *mahalle* for Istanbul). We do not predict the exact next place; these two properties are easier to learn and, for most uses, enough. Category is a small, fixed label set. Region is a large one: we frame next-region as classification, where each candidate census tract is one class, from about five hundred (Istanbul) to about eight thousand five hundred (California).

The motivation is practical: the category says what kind of place comes next, which indicates what a user will want; the region says where, which indicates where to prepare. Such preparation has support in the mobility literature, which has long built city services on location-based social network traces [34]. A recent analysis of tourist check-in mobility finds that visits concentrate on a few hub places, argues that crowds and demand therefore concentrate as well, and ties this structure to infrastructure planning and adaptive strategies [6]. That analysis reads the structure of past visits; we predict two properties of the next one, and aggregated over users those predictions point to where demand is heading before it lands. A census tract is a neighborhood, not a radio cell (the small coverage area a single base station serves), so we scope our claims to neighborhood-level preparation: demand and load anticipation, content staging, and capacity planning; not cell association or handover. We build and evaluate no such service here; Section VII quantifies what these predictions would give such a service.

IV. Method

A. The check-in-level representation

We want each visit, not each place, to carry its own vector. A standard place embedding gives every point of interest one fixed vector, so two visits to the same café look identical even when they happen at different times of day, in different company, or after different trips. We instead describe each check-in in its own context. Figure 1 shows the full data flow, from the raw check-in trail to the two outputs.

We build a graph with four levels: the check-in, the place it records, the region (a census tract) the place sits in, and the city. Edges tie these levels together and tie check-ins to one another: consecutive visits by the same user, weighted by how close in time they fall, and visits to the same place; nearby places are also linked at the place level. Each visit's category and time of day enter as input features of its node rather than as edges. This keeps the place-to-region-to-city geography of a hierarchical graph and adds the check-in as a new bottom level. We train the graph with an infomax objective, which contrasts real neighborhoods against shuffled ones, so each vector learns to match its true neighborhood and reject a fake one [1], [9]. The objective uses no task label: it never sees the next category or the next region. From the trained graph we read out one vector per check-in, and a model then sees a sequence of these per-visit vectors rather than a sequence of repeated per-place ones.

Each ingredient is individually standard: per-visit context, hierarchical place graphs, and the infomax objective all appear in prior work. The novelty is their combination, producing a vector per check-in from inside a hierarchical-graph-infomax representation, which earlier contextual embeddings built from sequence models (CTLE) do not provide. We show later that this combination, not extra supervision, is what makes the category task far easier to learn.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts two properties of the next visit at

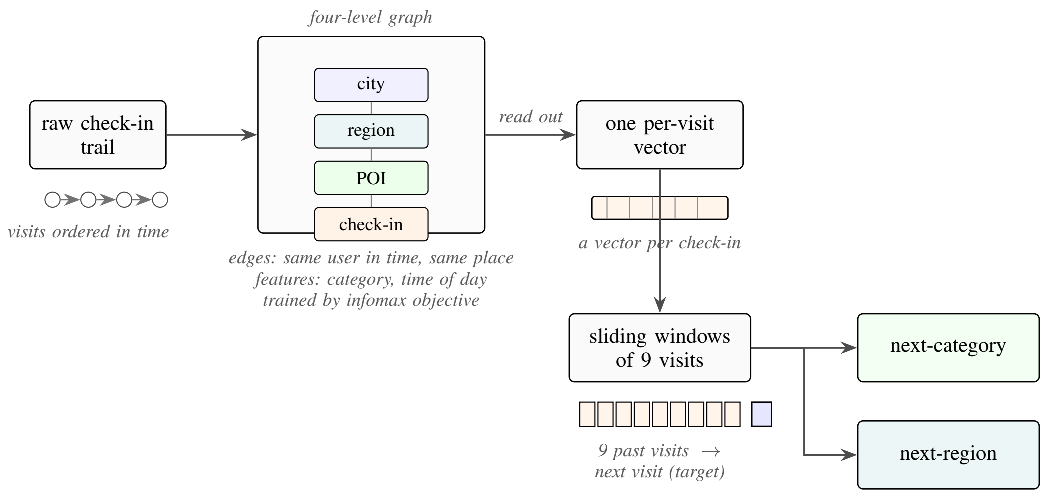

Fig. 1. From a check-in trail to two predictions: each visit is embedded in a four-level graph (check-in, place, region, city) and described in its own context; overlapping nine-visit windows feed a single model that outputs the next category and the next region.

once: its category and its region. Figure 2 shows the design. Two input streams pass through private per-task encoders (a small input network dedicated to each task, with no weights shared between them) into a shared trunk (the stack of layers both tasks use), where the two streams exchange information (a cross-attention layer lets each stream read the other's features). The trunk then feeds two outputs, one for the category and one for the region. The region output keeps a private spatial path of its own, a small branch that the category task does not touch.

The private spatial path is a task-specific branch inside the one model, not a second model, not a second representation, and not a route that swaps models per task. The whole system is a single model: one set of shared parameters and one forward pass produce both predictions. This is the deployment property we care about, since a service that wants both answers runs a single model once, instead of two separate dedicated single-task models.

The training loss is a fixed-weight sum of the two outputs, and both outputs use plain unweighted cross-entropy. The two tasks pull in opposite directions on this choice: the region metric counts every test visit once, so an unweighted loss matches it, whereas the category metric, macro-F1, counts every class equally, so one might expect a class-weighted loss to help. We tested class-weighting on both heads and it did not help: it lowered region accuracy and category macro-F1 rather than raising it. We therefore keep plain unweighted cross-entropy on both and balance the two tasks with a fixed-weight sum, the category output weighted 0.75 and the region output 0.25. This is ordinary hard parameter sharing [7], kept deliberately plain. Any improvement of the joint model over the dedicated single-task models then comes from the shared representation, not from an adaptive weighting scheme.

The cost of serving both tasks this way is modest. The joint model carries both task heads, so it is larger than a single dedicated model. It is still substantially cheaper to serve than the two separate dedicated models the usual setup would deploy: measured on our architecture, it needs about 17 to 20 percent fewer parameters and about 28 percent fewer operations per pair of predictions than the two dedicated models combined, so two answers come at well under the price of two.

V. Experimental Setup

A. Data

We evaluate on six datasets, summarized in Table I. Five come from Gowalla, a location-based social network, and cover the U.S. states of Alabama, Arizona, Florida, Texas, and California [2]; the sixth is the city of Istanbul, drawn from the Massive-STEPS collection [35], which gives us a non-U.S. check on the findings. We chose this spread on purpose. The states run from small to large, from about 114 thousand check-ins and 1,109 regions in Alabama to several million check-ins and 8,501 regions in California, so we can see whether a result holds as the number of regions grows. Every dataset uses the same seven place categories: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The region unit is a census tract for the Gowalla states and a mahalle (a municipal neighborhood) for Istanbul, and the number of regions ranges from about five hundred to about eight thousand five hundred.

B. Windows, splitting, and the integrity of the representation

Windows. For each user with at least ten visits we sort the check-ins in time and form overlapping windows of nine visits, one starting at each visit, with the next visit as the target. Overlapping windows give the category task more examples to learn from. Because the windows overlap, many of a user's windows end on the same final visit, over-representing that one target; we keep a single window ending on each user's last visit and drop the duplicates. The kept window is a full-context example, not padding.

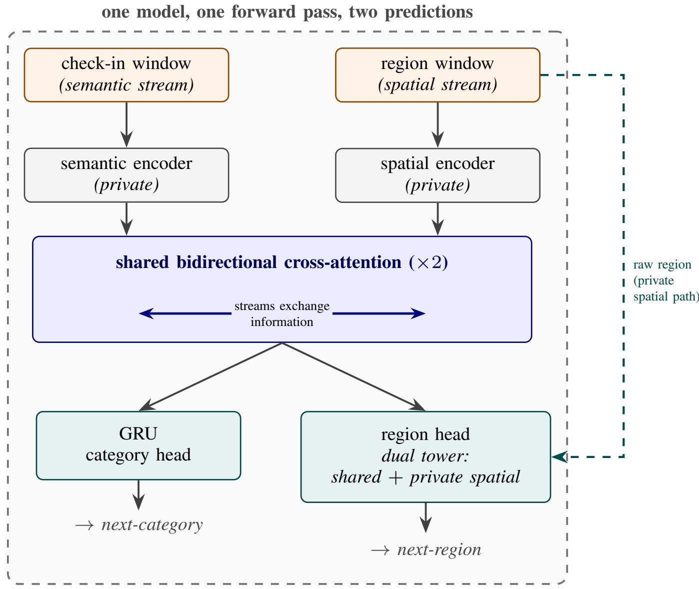

Fig. 2. The single multi-task model. A check-in (semantic) window and a region (spatial) window pass through private encoders into a shared cross-attention trunk, the only shared parameters; a sequence head predicts the next category, and the region head combines the shared output with a private spatial path. One model, one forward pass, two predictions.

Splitting. We split by user with stratified five-fold cross-validation, so all of a user's windows fall in the same fold. Overlapping windows therefore cannot leak across the split, because a test user's visits never appear in training. The split is stratified on the next-category label, which preserves the class proportions across folds, and this matters because the seven-class category task is imbalanced: Food is the majority class in every dataset, from about 25 percent of visits in Florida to 34 percent in Alabama.

Integrity of the representation. We train the representation once on the whole dataset and then feed it to both the dedicated single-task models and the joint model. It is therefore worth being explicit that it carries no information about the test visits. It does not, for three reasons. First, the representation is trained without any task label: its objective only contrasts real graph neighborhoods against shuffled ones, and it never sees the next-category or next-region target, so no label can travel from test to train through it. Second, the only thing it could carry from the test side is graph structure (which places sit near which, in space, time, and category), because the graph is built over all places. We measured that exposure directly: for each fold we rebuilt the representation from only that fold's training users and re-ran both tasks, comparing it against the full-corpus representation on the same folds. The effect is within fold noise and shows no systematic gain from the wider exposure: on region the change is -0.33 points at Alabama, $+0.01$ at Arizona, and -0.12 at Florida. On category it is $+0.29$ at Alabama, $+0.27$ at Arizona, and $+0.00$ at Florida, measured on the in-coverage visits whose places appear in training (about 67 percent at Alabama to 87 percent at Florida); this is a place-level proxy, since the representation cannot score a place never seen in training. Third, the one component that could pass visit-to-visit information is a region-transition prior: a table, built from training data, of how often visits move from one region to the next, used to bias region scores. An earlier version that used the whole dataset inflated region accuracy by 13 to 27 points across states, which is why any such table is built per fold, from training data only, seed by seed. The joint and dedicated models reported here run with this prior disabled entirely; it survives only in the HMT-GRN baseline (Section V-D), with the per-fold training-only build. Every learned baseline representation is pre-trained the same way, so all inputs are held to the same standard. The one residual we cannot fully measure is visits to places never seen in training: the representation is trained on all places, so it cannot score a place that is absent from training. On the visits we can measure, which are the large majority, the audit finds no inflation, and a planned training-only variant is meant to close the remaining gap.

C. *Metrics, superiority versus non-inferiority*

For the category task we report macro-averaged F1 (macro-F1), which averages the per-class F1 so that each of the seven categories counts equally, including the rare ones; its reference point is the majority-class floor, the macro-F1 of always predicting the single most common category. For the region task we report accuracy at ten (Acc@10), the fraction of test visits whose true region appears among the model's ten highest-scoring guesses; its comparison anchor is the dedicated single-task model.

We make two kinds of comparison and test each one appropriately. Where the joint model is meant to outperform, we test *superiority* with paired tests over matched runs: at the four datasets measured at four seeds, a paired test over the per-seed means with a Holm correction across the cells; at the two single-seed states, a paired Wilcoxon signed-rank test over the folds. Where the joint model is meant only to stay even, we test *non-inferiority*, that it is no worse than the dedicated model by more than a two-point margin, with a two-one-sided-tests (TOST) procedure. We fix the two-point margin in advance and on deployment grounds, before looking at whether any claim survives it. As in Section III, a mobility-aware service acts on which region will be busy, not on a single rank position. A two-point shift in Acc@10 is therefore below the granularity at which such a service would behave differently. The margin is also large relative to the task: with five hundred twenty to eight thousand five hundred regions, a random top-ten guess lands the true region at most about two percent of the time, so two points span a wide band of the achievable range. Finally, the equivalence is well powered. The standard deviation of the paired difference is small (0.04 to 0.15 points at the four multi-seed datasets, 0.07 at the two single-seed ones), which gives a power near 1.0 to reject a true two-point gap, so a non-inferiority verdict reflects a real match rather than a noisy test. Where the joint model outperforms and where it matches is a result, reported in Section VI, not here.

D. Baselines

We organize the comparisons into three roles, kept separate so each answers its own question. The first role is the state of the art for each task, on our data and folds. For the next-category task this is POI-RGNN [36], run as a faithful author implementation, and a simple "what usually follows what" Markov baseline over the nine recent categories (Markov-9). For the next-region task this is HMT-GRN [19], our primary region-native comparison, predicting the region from its own end-to-end recurrent model on the same data, seed, and folds; we run the HMT-GRN-style trunk and its region-transition prior, and drop its graph module and hierarchical beam search, which exist to prune a next-place head we do not predict. The HMT-GRN comparison is therefore a region-native model, not a component-complete reproduction. Alongside it we run a faithful STAN [14], trained from raw with its own embeddings (not ours) and its own sequence construction, under one audited recipe whose per-dataset precision and early-stopping variants are quality-neutral within a tenth of a point, and a ReHDM [37] reference reported under that method's own published protocol rather than re-run under our matched protocol. The second role is a representation control, run end to end on Florida and repeated in frozen form at Alabama, Arizona, and Istanbul, to attribute the category gain to our specific design and not to contextualization in general or to extra features: CTLE [10], the closest prior contextual check-in embedding, presented fairly in both its end-to-end and frozen forms, and a feature-concatenation control that appends raw per-visit features to a standard place embedding under the same model. The third role is a design comparison between the two ways of coupling the tasks. The dominant published coupling is the directed cascade, which predicts the category first and conditions the place prediction on it [25], [26], [28]; CSLSL [27], its current form, even reports the chain outperforming a parallel variant that shares one trunk, so the choice of coupling is not settled in advance. We test the cascade inside our own model rather than through a re-implementation of those systems: we remove the layer where the two streams exchange information and instead feed the predicted category forward into the region pathway, keeping the representation, the outputs, and the training recipe unchanged. This isolates what we actually choose, the coupling topology (chain versus parallel), at equal parameter and compute cost. The dedicated single-task models are the comparison anchors for the joint model throughout, and the standard place-level embedding [1] is the place-level comparison that isolates what the per-visit representation adds.

VI. Results

A. The representation makes the category task learnable

Read this as: describing each visit in its own context, rather than each place once, is what makes the next-category task learnable. Table II compares our check-in-level representation against a standard place embedding (HGI [1]) on next-category macro-F1, both fed to the same dedicated model under the windowing used throughout the paper. This is a controlled, like-for-like comparison: the target, the single-task head, the training recipe, and the folds are identical across the two columns, and only the input representation changes (a vector per visit versus a vector per place), so the margin isolates the representation, not a difference in task or model. (For this reason the check-in-level column keeps one fixed recipe and is intentionally not the per-dataset-tuned dedicated ceiling of Table III.) The check-in-level representation outperforms the place embedding by a wide and consistent margin at every state: $+29.31$ at Alabama, $+27.63$ at Arizona, $+39.62$ at Florida, $+37.95$ at California, and $+37.47$ at Texas. The same holds on Istanbul ($+28.09$). The gains fall in two bands, about $+28$ to $+29$ points at the smaller datasets (Istanbul, Alabama, Arizona) and about $+37$ to $+40$ at the three large states. what the per-visit representation reaches. A controlled test attributes most of that gain to the per-visit context itself: replacing each visit's vector with a per-place-averaged version of the same representation gives back only part of the margin. This leaves roughly 64 to 90 percent of the gain (state-dependent) to the context each visit carries, not to extra training signal.

Figure 3 shows why the representation helps this task in particular. By category, the per-visit vectors are cleanly grouped. Their silhouette score by category, a separation measure from -1 to 1 where higher means the seven labeled groups are tighter and better separated, is about 0.57, against about 0.00 for the place embedding. Their nearest-neighbor category purity, the share of a vector's nearest neighbors that carry its own category, is about 0.98, against about 0.78 (both averaged over the five U.S. states). The same geometry does not separate regions, which is exactly the point: the representation improves category separability but carries no spatial structure, so its benefit is category-only and sets up

TABLE I
DATASET STATISTICS, ORDERED BY NUMBER OF REGIONS: FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, FROM MASSIVE-STEPS). ALL DATASETS SHARE THE SAME SEVEN ROOT CATEGORIES; THE REGION UNIT IS A CENSUS TRACT (A MAHALLE FOR ISTANBUL). WINDOWS ARE THE OVERLAPPING NINE-VISIT INPUTS PLUS THE NEXT VISIT AS TARGET; SEQUENCE LENGTHS ARE PER-USER CHECK-IN COUNTS; DENSITY IS THE CHECK-IN-MATRIX FILL, $100 \times$ CHECK-INS/(USERS $\times$ POIS); MAJORITY IS THE SHARE OF VISITS IN THE MOST COMMON CATEGORY (FOOD IN EVERY DATASET).

Dataset	**Source**	**Check-ins**	**Users**	**POIs**	**Regions**	**Windows**	**Max len**	**Avg len**	**Density (%)**	**Majority (%)**
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

the joint study in Section VI-B. A direct comparison with the closest prior contextual embedding makes the source of the gain concrete. CTLE [10] also gives each visit its own vector, but it learns from place identifiers and timestamps alone: its pretraining recovers masked places and masked hours, so the category vocabulary never enters its training, whereas our graph reads each visit's category and time of day as input features (and even the place embedding builds its place vectors from category-derived features). Fine-tuned end to end at Florida, at its authors' architecture defaults scaled to the shared 64-dimension budget and under the same windowing, CTLE reaches 33.45 macro-F1 at its best epoch (29.69 at its final epoch), about two points below the place embedding scored under the same best-epoch rule and far below our 75.15; frozen and fed to the same single-task model as ours, the same control at Alabama, Arizona, and Istanbul repeats the ordering, with margins of $+37.8$, $+37.0$, and $+28.7$ macro-F1. A feature-concat control, the place embedding joined with raw per-visit features and fed to the same model, does not close the gap either, so the category gain comes from the hierarchical per-visit representation, not from contextualization alone and not from feature injection.

B. One model, two tasks

Read this as: one model outperforms a dedicated next-category model at every dataset and outperforms or matches a dedicated next-region model, with the region gain rising with the number of regions across the U.S. states. To calibrate the two scales: category macro-F1 is taken over seven classes (a majority-class floor of about 7%), and region Acc@10 is taken over hundreds to thousands of regions, where a random top-ten guess is right at most about two percent of the time (and well under one percent at the larger region counts). Table III reports the single joint model against the dedicated single-task ceilings (we call them ceilings because a jointly trained head is usually expected to match or fall short of a head trained for that task alone). Each dedicated category ceiling is tuned per dataset over batch size and learning rate (the region ceilings use the strongest fixed recipe), so the joint model is compared against the strongest dedicated setting, not a convenient one. Figure 4 plots the signed gains ordered by region count. On category the joint model outperforms the dedicated ceiling at every dataset, by $+5.34$ to $+9.40$ macro-F1 (smallest at Florida, largest at Arizona, and $+8.59$ at Istanbul). On region, measured by Acc@10, the joint model outperforms the dedicated ceiling at Florida $+0.72$ (4,703 regions), Texas $+2.12$ (6,553), California $+2.20$ (8,501), and Istanbul $+0.28$ (520), and stays a non-inferior match (TOST, ± 2 pp) at Alabama (-0.31) and Arizona ($+0.10$). Across the five U.S. states the region gain rises with the number of regions, from -0.31 at the smallest to $+2.20$ at the largest, so the joint model helps the spatial task more, not less, as the number of regions grows; the state with the most regions has the largest region gain, the opposite of a cost that grows with the size of the spatial problem. We order by region count because that is the axis of interest; region count and corpus size co-vary across these datasets, so we read the trend across the points rather than as a precise law.

Four of the six datasets are measured at $n{=}20$ on both arms: Alabama, Arizona, Florida, and Istanbul average four seeds over the same five frozen folds, and the tests pair the per-seed means (four paired observations per dataset). On category, all four seeds favor the joint model at every one of these datasets, and each per-dataset gain is significant after a Holm correction across the four cells (paired t, corrected $p < 10^{-6}$ everywhere). On region, all four are tested equivalences (TOST, ± 2 pp), and all four pass with 90% confidence intervals well inside two points: Alabama (-0.31; -0.48 to -0.14), Arizona ($+0.10$; $+0.00$ to $+0.21$), Florida ($+0.72$; $+0.67$ to $+0.78$), and Istanbul ($+0.28$; $+0.24$ to $+0.32$). At Florida and Istanbul the whole interval sits above zero, so the joint model outperforms there as well; at Arizona the interval touches zero, so we report a match, not a gain. California and Texas remain provisional: their dedicated ceilings are $n{=}20$, but the joint cells are a single seed over five folds. On those folds, all five favor the joint model on both tasks (one-sided paired Wilcoxon at its $n{=}5$ floor, $p{=}0.031$), and the region gains carry 90% intervals entirely above the two-point margin ($+2.14$ to $+2.28$ at California, $+2.04$ to $+2.18$ at Texas). A multi-seed run at these two states is the remaining confirmation.

A reader used to multi-task learning [7] expects the harder task to pay for the easier one, so it is worth saying where the category gain comes from. A controlled test isolates the source: we freeze the region pathway at the start of training so it cannot learn, and therefore cannot teach the category task, yet the full category lift survives at Alabama, Arizona, and Florida (within 0.3 of the joint model and far above the dedicated single-task model). We therefore read the category gain as a stronger shared trunk, obtained at no extra deployment cost (one model, one forward pass), rather than the region task teaching the category one; we report this as a finding, not a hypothesis.

The joint model also stands above every external baseline on both tasks, at every dataset where each is reported (Table III): the primary region-native comparison, HMT-GRN [19] (for example California 49.61 versus our 65.69), a from-raw STAN [14] and a ReHDM reference [37] on region, and POI-RGNN [36] and a Markov floor on category, all trail it.

We also test our choice of coupling the tasks in parallel rather than in a chain. We prefer the parallel form on principle: the two targets are co-equal, so a chain would have to pick one to predict first and pass its errors to the other. The empirical record still had to be checked, because the cascade dominates published practice, and CSLSL reports its chain outperforming a parallel variant that shares one trunk on its own benchmarks [27]. Rewired as a cascade (Section V-D), our model reaches the same combined accuracy as its parallel form at the same parameter and compute cost (a joint-score difference of about zero at Alabama, Arizona, Florida, and Istanbul, the four datasets where we ran the comparison, spanning 520 to 4,703 regions and including the non-U.S. city). We read this as a defense of the parallel design, not a claim that we outperform the cascade: on this representation the coupling topology makes no measurable difference in either direction, and the simpler parallel structure loses nothing. Two qualifications bound this reading. The cascade variant runs under the training recipe tuned for the parallel model, and its coupling form was fixed in advance rather than searched, so a cascade tuned as carefully as the parallel design remains future work.

C. *External validity (Istanbul)*

Read this as: the finding holds on a non-U.S. city, measured on the same representation, protocol, and sample size as the U.S. states. On Istanbul the joint model outperforms the dedicated category ceiling by $+8.59$ macro-F1 (54.74 to 63.33) and edges past the dedicated region ceiling at $+0.28$ Acc@10 (75.16 to 75.44), a statistically supported gain (the 90% confidence interval of the paired difference lies entirely above zero; it also passes the non-inferiority test, TOST, ± 2 pp); both arms average four seeds (n=20). We report Istanbul as a lift over the dedicated ceiling, not as an absolute Acc@10, because its region count (520 mahalle) differs from the Gowalla states. The representation, the overlapping-window protocol, and the training setup are identical to those used for the Gowalla states, so the comparison is like-for-like. The external baselines line up the same way as on the U.S. states: on category the joint model is far above a nine-step Markov floor (24.55) and POI-RGNN [36] (30.12), and on region it is above ReHDM [37] (69.33), a faithful STAN [14] (61.86), and HMT-GRN [19] (60.4). The picture from the U.S. states repeats on a different continent and a different region unit, and on region Istanbul comes out slightly ahead rather than merely close.

TABLE II
NEXT-CATEGORY MACRO-F1 OF THE CHECK-IN-LEVEL REPRESENTATION AGAINST A STANDARD PLACE EMBEDDING (HGI), SAME SINGLE-TASK HEAD, RECIPE, AND FOLDS (MEAN AND FOLD STANDARD DEVIATION, SEED 0).

Dataset	Check-in level	Place level	Δ
Istanbul	**54.65** $\pm$0.6	26.56 $\pm$0.5	+28.09
AL	**55.87** $\pm$2.4	26.56 $\pm$1.0	+29.31
AZ	**57.13** $\pm$1.3	29.50 $\pm$1.0	+27.63
FL	**75.15** $\pm$0.5	35.53 $\pm$0.3	+39.62
TX	**69.95** $\pm$0.2	32.48 $\pm$0.6	+37.47
CA	**70.26** $\pm$0.3	32.31 $\pm$0.4	+37.95

The matching place-level value at Alabama and Istanbul (26.56) is a coincidence of two independent runs; their per-fold means differ.

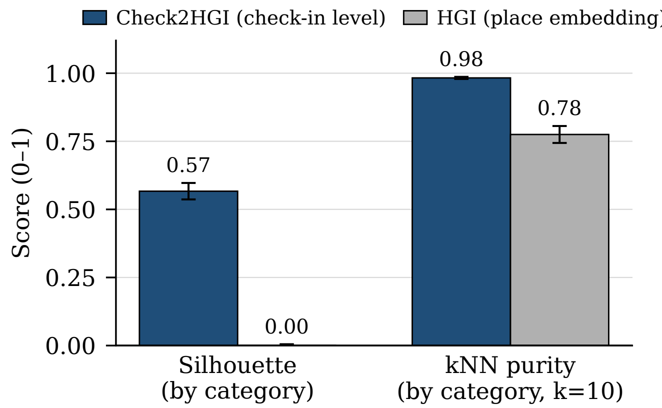

Fig. 3. Class separability of the representations on the ground-truth next-category labels (cosine silhouette; leave-one-out nearest-neighbor (kNN) purity, k=10), averaged over the five U.S. states. The check-in-level representation separates the seven categories; the place embedding does not.

VII. Discussion and Limitations

The clearest reading of our results is a design one: one model is enough. It pairs a shared trunk that carries the semantic context with a private spatial path that keeps next-region accuracy from degrading. The shared context lifts the next-category task sharply over a dedicated model at every dataset, and the private path keeps the next-region task competitive: a non-inferior match (TOST, ± 2 pp) at the two small U.S. states, and a tested gain at the other four datasets. Across the U.S. states the region gain rises with the number of regions, growing rather than fading (Figure 4). Overall, the joint model matches or outperforms the dedicated single-task model on both tasks at once (Table III). For a mobility-aware service, this points to a practical option: one model, one forward pass, can anticipate both what a user will do next and where, without a second model and without a task trade-off.

A short usage sketch makes the motivation concrete, with numbers rather than adjectives. A service could treat the model's top-ranked next regions as an anticipatory set: at California, a shortlist of ten regions out of 8,501 candidates, about a tenth of a percent of the space, contains the true next region 65.69 percent of the time, more than five hundred times

TABLE III

ONE MODEL, TWO TASKS: THE SINGLE JOINT MODEL AGAINST THE DEDICATED SINGLE-TASK MODELS AND THE EXTERNAL BASELINES, ORDERED BY REGION COUNT. CATEGORY IS MACRO-F1, REGION IS ACC@10 (MEAN AND STANDARD DEVIATION ACROSS RUNS). BOLD MARKS A STATISTICALLY SUPPORTED IMPROVEMENT OVER THE DEDICATED MODEL; IN THE REGION COLUMNS, $\uparrow$ MARKS THAT SUPPORTED IMPROVEMENT AND $\approx$ A NON-INFERIOR MATCH (TOST, ± 2 PP; SECTION VI-B).

		Next-category (macro-F1)				Next-region (Acc@10)				
Dataset	**Regions**	**Markov-9**	**POI-RGNN**	**Dedicated**	**Joint (ours)**	**HMT-GRN**	**ReHDM**	**STAN**	**Dedicated**	**Joint (ours)**
Istanbul	520	24.55	30.12	54.74 ± 0.09	**63.33** ± 0.02	60.4	69.33	61.86	75.16 ± 0.01	**75.44** $\pm 0.04^{\uparrow}$
AL	1,109	20.50	23.80	56.82 ± 0.03	**64.54** ± 0.10	57.05	65.38	60.72	70.11 ± 0.10	69.80 $\pm 0.05^{\approx}$
AZ	1,547	23.92	27.64	56.43 ± 0.10	**65.84** ± 0.02	43.70	53.00	49.86	59.46 ± 0.08	59.56 $\pm 0.05^{\approx}$
FL	4,703	29.74	34.49	74.51 ± 0.03	**79.85** ± 0.03	63.74	64.49	72.99	76.70 ± 0.02	**77.42** $\pm 0.03^{\uparrow}$
TX	6,553	28.67	33.03	69.79 ± 0.08	**77.23** ± 0.12	53.85	48.81‡	61.67†	64.95 ± 0.01	**67.07** $\pm 0.45^{\uparrow}$
CA	8,501	27.58	31.78	70.60 ± 0.07	**77.04** ± 0.20	49.61	50.26‡	58.52†	63.49 ± 0.02	**65.69** $\pm 0.30^{\uparrow}$

HMT-GRN (region-native, on our folds) is the primary external comparison; STAN (faithful, from raw) and ReHDM (its own protocol) are references. †STAN partial-fold cells: Texas four of five folds, California two of five (seed 0). ‡ReHDM at Texas and California is a single seed. The joint cells at California and Texas are a single seed over five folds (provisional); every other cell in our columns averages four seeds. Standard deviations are across seeds for four-seed cells and across folds for single-seed cells.

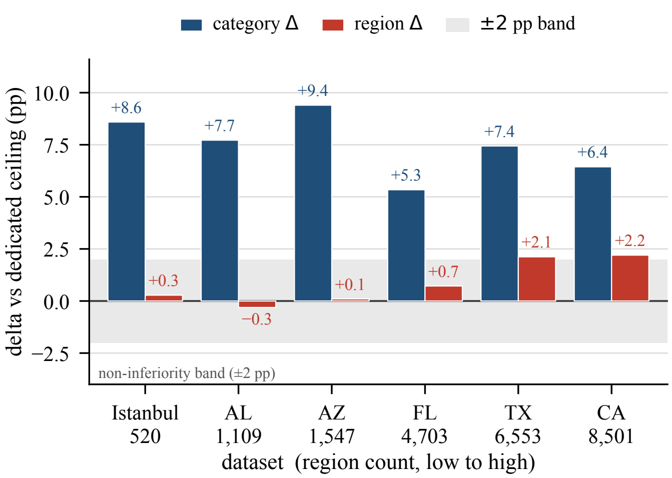

Fig. 4. Signed gains of the joint model over the dedicated single-task models, ordered by region count. The category gain (macro-F1) is positive at every dataset; the region gain (Acc@10) rises with region count across the five U.S. states and is also positive at Istanbul. The two metrics are on different scales; the ± 2-point band applies to region.

better than picking ten at random; at Alabama, ten out of 1,109 capture it 69.80 percent of the time. The predictions are also geographically concentrated. On the four datasets where we measured it (Alabama, Arizona, Florida, and Istanbul; a single seed over five folds), the ten shortlisted regions sit, on average, about 3 to 8 kilometers from the shortlist's own geographic center (the median across visits, computed from region centroids). The miss distance is similar: when the top guess is wrong and the true region was seen in training, the predicted region's centroid sits a median of about 3 to 8 kilometers from the true region's, while two randomly drawn regions from the same map sit a median of 20 to 241 kilometers apart; visits to regions never seen in training are rarer and miss by more. The next category then hints at what to prepare there. This remains motivation, not a measured service result: we report properties of the model's own predictions, not a system's cost or a service's outcome.

Three limits qualify these results. First, the joint numbers at California and Texas come from a single seed over five folds, so those two cells are provisional; the other four datasets average four seeds on both arms. A multi-seed run at the two largest states is the remaining step. Second, our representation is trained once over all places. One slice of its behavior, visits to places never seen during training, is the single effect we cannot fully isolate. The same property sets the deployment path: the representation is a precomputed artifact, so a served system would embed new check-ins by refreshing it on a schedule, as embedding infrastructures commonly do, rather than per visit; an encoder that embeds unseen visits directly is the same machinery as the planned variant that trains on each fold's training places only, and both are one study. Third, although a mobility-aware service is our motivation throughout, we do not build or evaluate one; a concrete service evaluation, with its own baselines and costs, remains the natural next study.

Finally, our representation is a fixed per-visit vector, computed once and left unchanged; any downstream model can consume it. Plugging it into other sequence models, and applying it to mobility tasks beyond next-category and next-region, are natural directions to explore. Structural analysis of check-in networks has itself named machine learning as a next step [6]; learning over check-in structure, as this study does, moves in that direction.

VIII. CONCLUSION

We asked whether one model can predict both the category and the region of a user's next visit. Two findings answer it. First, a check-in-level representation, where each visit gets its own vector instead of one fixed vector per place, makes the next-category task far more learnable than the standard place-level representation does. Second, on top of that representation, a single multi-task model outperforms a dedicated category model at every dataset, and on region it does so at four of the six datasets while remaining non-inferior (TOST, ± 2 pp) at the other two. In summary, one model that shares a semantic context across both tasks while keeping a private spatial path for region can anticipate both what a user will do next and where, without giving up the next-region task. Across the U.S. states the gain on region grows with the number of regions. We will release the code for the model, the representation, the baselines, and the statistical tests upon publication; both data sources are already public: the Gowalla check-ins [2] and the Massive-STEPS collection [35].

REFERENCES

[1] W. Huang, D. Zhang, G. Mai, X. Guo, and L. Cui, "Learning urban region representations with POIs and hierarchical graph infomax," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 196, pp. 134–145, 2023.

[2] E. Cho, S. A. Myers, and J. Leskovec, "Friendship and mobility: User movement in location-based social networks," in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, 2011, pp. 1082–1090.

[3] C. Song, Z. Qu, N. Blumm, and A.-L. Barabási, "Limits of predictability in human mobility," *Science*, vol. 327, no. 5968, pp. 1018–1021, 2010.

[4] E. Bastug, M. Bennis, and M. Debbah, "Living on the edge: The role of proactive caching in 5G wireless networks," *IEEE Communications Magazine*, vol. 52, no. 8, pp. 82–89, 2014.

[5] C. L. Vielhaus, J. V. S. Busch, P. Geuer, A. Palaios, J. Rischke, D. F. Külzer, V. Latzko, and F. H. P. Fitzek, "Handover predictions as an enabler for anticipatory service adaptations in next-generation cellular networks," in *Proceedings of the 20th ACM International Symposium on Mobility Management and Wireless Access (MobiWac)*. ACM, 2022, pp. 19–27.

[6] D. L. L. Moura, A. L. L. Aquino, and A. A. F. Loureiro, "On the design of mobility-aware systems: A tourist's perspective," in *2025 International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, 2025, pp. 667–674.

[7] R. Caruana, "Multitask learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[8] V. H. O. Silva, I. F. Almeida, T. S. Paiva, G. B. Santos, F. A. Silva, and F. T. Sousa, "An investigation into multi-task learning for point-of-interest category classification and next-POI prediction," in *Anais do Congresso Brasileiro de Inteligência Computacional (CBIC)*, 2025.

[9] P. Veličković, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, "Deep graph infomax," in *Proceedings of the 7th International Conference on Learning Representations (ICLR)*, 2019.

[10] Y. Lin, H. Wan, S. Guo, and Y. Lin, "Pre-training context and time aware location embeddings from spatial-temporal trajectories for user next location prediction," in *Proceedings of the 35th AAAI Conference on Artificial Intelligence*, vol. 35, 2021, pp. 4241–4248.

[11] Q. Liu, S. Wu, L. Wang, and T. Tan, "Predicting the next location: A recurrent model with spatial and temporal contexts," in *Proceedings of the 30th AAAI Conference on Artificial Intelligence*, 2016, pp. 194–200.

[12] J. Feng, Y. Li, C. Zhang, F. Sun, F. Meng, A. Guo, and D. Jin, "DeepMove: Predicting human mobility with attentional recurrent networks," in *Proceedings of the 2018 World Wide Web Conference (WWW)*, 2018, pp. 1459–1468.

[13] D. Yang, B. Fankhauser, P. Rosso, and P. Cudré-Mauroux, "Location prediction over sparse user mobility traces using RNNs: Flashback in hidden states!" in *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020, pp. 2184–2190.

[14] Y. Luo, Q. Liu, and Z. Liu, "STAN: Spatio-temporal attention network for next location recommendation," in *Proceedings of the Web Conference 2021 (WWW)*, 2021, pp. 2177–2185.

[15] S. Yang, J. Liu, and K. Zhao, "GETNext: Trajectory flow map enhanced transformer for next POI recommendation," in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022, pp. 1144–1153.

[16] M. Luca, G. Barlacchi, B. Lepri, and L. Pappalardo, "A survey on deep learning for human mobility," *ACM Computing Surveys*, vol. 55, no. 1, pp. 7:1–7:44, 2021.

[17] Y. Li, T. Chen, Y. Luo, H. Yin, and Z. Huang, "Discovering collaborative signals for next POI recommendation with iterative Seq2Graph augmentation," in *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 1491–1497.

[18] Z. Sun, Y. Lei, L. Zhang, C. Li, Y.-S. Ong, and J. Zhang, "A multi-channel next POI recommendation framework with multi-granularity check-in signals," *ACM Transactions on Information Systems*, vol. 42, no. 1, pp. 1–28, 2024.

[19] N. Lim, B. Hooi, S. Ng, Y. L. Goh, R. Weng, and R. Tan, "Hierarchical multi-task graph recurrent network for next POI recommendation," in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022, pp. 1133–1143.

[20] N. Zhu, J. Cao, X. Lu, C. Liu, H. Liu, Y. Li, X. Luo, and H. Xiong, "Predicting a person's next activity region with a dynamic region-relation-aware graph neural network," *ACM Transactions on Knowledge Discovery from Data*, vol. 16, no. 6, pp. 1–23, 2022.

[21] K. Sun and M. Xu, "Knowledge graph tokenization for behavior-aware generative next POI recommendation," arXiv:2509.12350, 2025, arXiv preprint.

[22] D. Liao, W. Liu, Y. Zhong, J. Li, and G. Wang, "Predicting activity and location with multi-task context aware recurrent neural network," in *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, 2018, pp. 3435–3441.

[23] L. Zhang, Z. Sun, J. Zhang, Y. Lei, C. Li, Z. Wu, H. Kloeden, and F. Klanner, "An interactive multi-task learning framework for next POI recommendation with uncertain check-ins," in *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020, pp. 3551–3557.

[24] Y. Wang, H. Chen, S. Liu, J. Zhang, L. Wu, X. Cui, and Y. Hu, "Hierarchy aware-based multi-task learning for user location prediction," *The Journal of Supercomputing*, vol. 81, no. 11, 2025.

[25] J. Ye, Z. Zhu, and H. Cheng, "What's your next move: User activity prediction in location-based social networks," in *Proceedings of the 2013 SIAM International Conference on Data Mining (SDM)*, 2013, pp. 171–179.

[26] J. He, X. Li, and L. Liao, "Category-aware next point-of-interest recommendation via listwise Bayesian personalized ranking," in *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI-17)*, 2017, pp. 1837–1843.

[27] Z. Huang, S. Xu, M. Wang, H. Wu, Y. Xu, and Y. Jin, "Human mobility prediction with causal and spatial-constrained multi-task network," *EPJ Data Science*, vol. 13, no. 1, p. 22, 2024.

[28] F. Yu, L. Cui, W. Guo, X. Lu, Q. Li, and H. Lu, "A category-aware deep model for successive POI recommendation on sparse check-in data," in *Proceedings of The Web Conference 2020 (WWW)*, 2020, pp. 1264–1274.

[29] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, "Gradient surgery for multi-task learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.

[30] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, "GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks," in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018, pp. 794–803.

[31] A. Navon, A. Shamsian, I. Achituve, H. Maron, K. Kawaguchi, G. Chechik, and E. Fetaya, "Multi-task learning as a bargaining game," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022, pp. 16 428–16 446.

[32] D. Xin, B. Ghorbani, A. Garg, O. Firat, and J. Gilmer, "Do current multi-task optimization methods in deep learning even help?" in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[33] V. Kurin, A. D. Palma, I. Kostrikov, S. Whiteson, and M. P. Kumar, "In defense of the unitary scalarization for deep multi-task learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[34] T. H. Silva, A. C. Viana, F. Benevenuto, L. Villas, J. Salles, A. A. F. Loureiro, and D. Quercia, "Urban computing leveraging location-based social network data: A survey," *ACM Computing Surveys*, vol. 52, no. 1, pp. 17:1–17:39, 2019.

[35] W. Wongso, H. Xue, and F. D. Salim, "Massive-steps: Massive semantic trajectories for understanding POI check-ins — dataset and benchmarks," arXiv:2505.11239, 2025, arXiv preprint.

[36] C. G. S. Capanema, G. S. de Oliveira, F. A. Silva, T. R. de Moura Braga Silva, and A. A. F. Loureiro, "Combining recurrent and graph neural networks to predict the next place's category," *Ad Hoc Networks*, vol. 138, p. 103016, 2023.

[37] X. Li, Z. Gu, R. Yao, Y. Zhou, H. Zhu, J. Zhao, and W. Du, "Beyond individual and point: Next POI recommendation via region-aware dynamic hypergraph with dual-level modeling," in *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, 2025, pp. 3081–3089.